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TEST REPORT

Lucideon Reference: N26458 (QT-78966/4/KNA)/Ref. 4

Project Title: Compressive Testing to BS EN 1052-1:1999 of L10 Low Carbon Bricks

Client: Earth4Earth Technology Ltd
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Opinions and interpretations expressed herein are outside the scope of UKAS Accreditation.

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1 INTRODUCTION

Earth4Earth Technology Ltd required compressive strength testing to be carried out on their L10 Low Carbon Bick bricks in accordance with BS EN 1052-1:1999 – Methods of Test for Masonry - Determination of Compressive Strength.

Testing was carried out on 5 May 2026 at Lucideon’s Structures Laboratory at Queens Road, Penkhull, Stoke-on-Trent, ST4 7LQ.

2 SAMPLES RECEIVED

Solid bricks referenced as L10 Low Carbon Bick were received from the client. The bricks were nominally 215 mm x 102.5 mm x 65 mm (length x width x height).

The bricks were solid with a tested compressive strength of 50.9 N/mm².

3 TEST PROGRAMME

3.1 Compression Tests to BS EN 1052-1:1999

Five specimens of N10 Carbon Negative bricks were constructed using M4 mortar.

4 METHOD OF CONSTRUCTION

5 No. specimens of L10 Carbon Negative bricks were constructed 5 No. courses high by 2 No. units wide in accordance with BS EN 1052-1.

The specimens were constructed on flat steel plates, which were supported on a steel clamp base.

The specimens were covered with polythene and cured for 28 days.

Specimens were constructed on 7 April 2026.

The compressive strength of the mortar was 4.09 N/mm^2.

5 SPECIMENS PREPARATION FOR INSTRUMENTATION

The five specimens constructed for compressive testing were instrumented to enable strains to be measured so that the modulus of elasticity could be calculated.

The strains were measured using a Demec Gauge (which manually locates onto two pre−installed stainless steel Demec Points at a set distance).

The Demec Points are approximately 5 mm in diameter and incorporate a small, indented locating hole.

The Demec Gauge used employs variable and fixed ends for location onto two Demec Points.

The gauge length (the distance between the points) was 150 mm.

Plate 1 shows the Demec location positions.

6 TEST PROCEDURE

All specimens were tested using a servo controlled hydraulic loading machine (Dartec).

A 'loading ramp' was set up to load the specimens to follow the prescribed rate in the Standard until failure.

The load was paused at increments to enable strains to be recorded manually as previously detailed.

Testing was carried out at a temperature of 20ºC and a relative humidity of 57%.

7 RESULTS

Individual compressive load results, strength results and mean values are given in Table 1.

Individual modulus of elasticity for the compressive specimens together with a mean value are given in Table 2.

Stress strain curves can be seen in Chart 1.

Characteristic strength is either the mean strength/1.2 or the lowest strength value ($f_{i\ min}$), whichever is the smaller, Clause 10.2(a) of BS EN 1052-1:1999.

Therefore, 7.53 N/mm^2 divided by 1.2 = 6.28 N/mm^2 and the lowest strength value was 6.78 N/mm^2.

As there were five or more samples Clause 10.2 of BS EN 1052-1:1999 requires the 5% fractile values to be calculated using a confidence level of 95%. This was calculated to be 5.69 N/mm^2.

Therefore, the characteristic compressive strength is the greater from Clause 10.2 (a) and 10.2, (b), which is 6.28 N/mm^2.

Plate 2 shows a general view of failure for the specimens.

TABLES

Table 1 – Mortar Properties

*Compressive Strength to EN 1015-11 (N/mm^2) *	*Air Content to EN 1015-7 (%)	*Flow Value to BS 4551
4.09	4.9	175

*These tests are not on Lucideon's UKAS Schedule.

Table 2 – Individual Compressive Results on Samples Constructed Using L10 Low Carbon Bick Bricks

Panel No.	Length (mm)	Width (mm)	Height (mm)	Failure Load (kN)	Compressive Strength at First Crack (N/mm^2)	Approximate Time to Fail Sample (Minutes)	Compressive Strength (N/mm^2)	Failure Mode
1	440	105	380	345	N/A	19	7.47	Catastrophic compressive failure
2	445	100	380	320	6.82	17	7.19	
3	440	100	380	390	8.65	22	8.86	
4	444	100	382	301	6.70	16	6.78	
5	444	101	380	330	7.32	18	7.36	
Mean	**442.6**	**101.2**	**380.4**	**337.2**	**-**	**-**	**7.53**	**-**

Table 3 – Individual Modulus of Elasticity Results Calculated from Compression Tests on L10 Low Carbon Bick Bricks

Panel No.	Maximum Compressive Load (kN)	Compressive Strength (N/mm^2)	Modulus of Elasticity (N/mm^2)
1	345	7.47	5555
2	320	7.19	8277
3	390	8.86	7496
4	301	6.78	8108
5	330	7.36	4481
Mean	**337.2**	**7.53**	**6783**

NOTE: The results given in this report apply only to the samples that have been tested.

END OF REPORT

PLATES

Plate 1 – General View of Set-Up Including Positions of Demec Points

Plate 2 – Specimen Following Catastrophic Compressive Failure

Chart 1 - Stress v Strain for Specimens Tested Under Compressive Load to BS EN 1052-1:1999